Cooperative Robots for Smart Factories

An open-source logistics integration blueprint

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Abstract

A TurtleBot 4 and a MyCobot 280 share logistics tasks in a didactic Siemens smart factory. The system links robot navigation, vision and manipulation through open-source software, using only onboard sensors.

The integration challenge

Robot platforms and factory PLCs use different software stacks. Reliable operation requires explicit
coordination across ROS 2 Jazzy, ROS 2 Galactic and the Siemens industrial network.

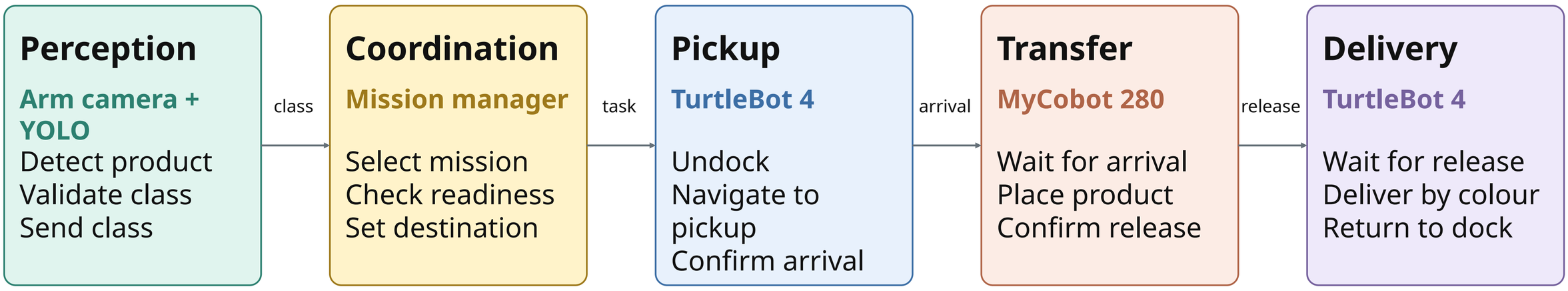
A modular solution

Separate interfaces connect the mobile base, manipulator and OPC UA factory gateway. The dashboard supervises shared missions, while each robot can also run independently.

Cooperative delivery pipeline

SLAM mapping · AMCL localisation · Nav2 navigation

MoveIt 2 motion planning + suction gripper



The arm waits for the mobile base to reach pickup. The base waits for product release before delivery.

Three logistics missions

Product delivery

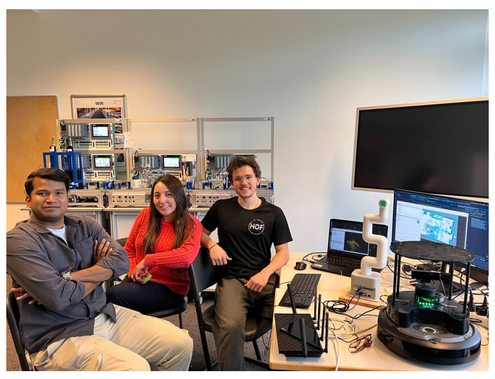
Collect the identified product, deliver it to the destination for its colour and return to dock.

Defective-part retrieval

Collect rejected components from the factory and transport them to the inspection area.

Material restocking

Collect supplies and bring them to the factory inventory station before returning to dock.





B002 laboratory and project team.

Evaluation & outlook

The next benchmark covers clear paths, static obstacles and moving obstacles. ROS bag recordings capture robot motion and mission events for cycle-time and completion-rate analysis.

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